LINH LAM

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PROFESSIONAL SUMMARY

Engineering physicist with hands-on experience in surface science, thin-film deposition, nanomaterials, and nanoscale characterization using AFM, SEM/EDX, and UV-Vis. Skilled in operating ultra-high vacuum (UHV) systems, cryogenic workflows, and precision laboratory instrumentation. Brings a strong foundation in experimental design, data analysis, troubleshooting, and technical reporting developed across photonics manufacturing, microelectronics workflows, and academic research. Highly motivated to contribute to atomically precise manufacturing and cutting-edge scanning probe microscopy research.

CORE COMPETENCIES

Semiconductor & Materials Engineering
• Thin-film coatings (optical & metal films)
• Nanowires, nanotubes, surface functionalization/passivation
• GaP, Si, Ni electrodeposition

Characterization Tools
• AFM, SEM/EDX, UV-Vis spectroscopy
• Profilometry, MS, GC, attended TEM and FIB

Process & Manufacturing Skills
• Thin-film process optimization
• Cleanroom/laboratory procedures
• Quality control, documentation, 5S, Lean

Technical Skillset
• Experimental design and data analysis
• UHV equipment
• Python, NumPy (Basic)
• Engineering communication & reporting

ENGINEERING & TECHNICAL EXPERIENCE

Research Assistant

McMaster University | Hamilton, ON | **2024 – 2025**
• Developed a conformal Ni electrodeposition method to uniformly coat individual GaP nanowires for betavoltaic device applications.
• Conducted SEM/EDX to evaluate coating quality.
• Performed HF and HCl etching in a **cleanroom** environment to remove native oxide from Si, GaAs, and GaP substrates.

• Performed routine handling of cryogens (liquid nitrogen) following laboratory SOPs.
• Attended TEM/FIB characterization sessions and learned fundamentals of image interpretation.

Manufacturing Technologist

Iridian Spectral Technologies | Ottawa, ON | **2023**
• Operated optical coating equipment using ultra high vacuum (UHV) machines.
• Performed routine operation of UHV chambers, pumps, and coating system maintenance.
• Prepared clear technical documentation, experimental logs, and summaries for engineering and research teams.
• Performed in-process measurements and quality control using optical metrology tools.

Research Assistant (Barry Lab)

Carleton University | Ottawa, ON | **2021-2022**

• Deposited a polymeric precursor on silicon wafer and glass using Atomic Layer Deposition.
• Monitored mass change during depositing process using in-situ quartz crystal microbalance and mass spectrometry.
• Handled liquid nitrogen and N_2O to supply carrier gases for ALD
• Characterized thin firm’s surface using AFM and SEM/EDX.

Undergraduate Research Intern (I-CUREUS Program)

Carleton University | Ottawa, ON | **2021**
• Synthesized **Ag nanocubes** via polyol method for plasmonic applications.
• Performed UV-Vis spectroscopy and **AFM (tapping mode)** surface characterization.
• Supported substrate preparation and functionalization processes.

EDUCATION

M.A.Sc. Engineering Physics, McMaster University, Hamilton, ON *(2024 – 2025)*
B.Sc. (Honours) Nanoscience, Minor in Mathematics, Carleton University, Ottawa, ON *(2018 – 2022)*

CERTIFICATE

• Workplace Hazardous Materials Information System (WHMIS) – McMaster University
• Controlled Goods Program (CGP) – Iridian Spectral Technologies (IDEX)
• Obtained Government of Canada Reliability (Level I) clearance – ECCC